

Fuzzy PID Design of Vehicle Attitude Control Systems

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Abstract: Movement of vehicle is susceptible to coupling, nonlinear and wind disturbance, which leads to violent changes of aerodynamics parameters. This paper provides a fuzzy PID design of vehicle attitude control systems to reduce the effect of random interference and uncertainty on the motion of the vehicle. The fuzzy relations between three PID control parameters and the deviation signal are used to adjust the PID control parameters online, the deviations of vehicle attitude control systems are reduced rapidly and effectively in complex and changeable environment. The simulation results prove that fuzzy-PID control can effectively accelerate the response speed, and enhance the dynamic performance of the vehicle control system.

Key Words: fuzzy control, fuzzy PID control, vehicle control system

1 INTRODUCTION

Movement of the vehicle in the atmosphere has the characteristics, such as nonlinearity, strong coupling, uncertainty and random disturbance, which leads to violent changes of aerodynamics parameters^[1]. In order to overcome the inaccuracy of missile model and random factors in the process of flight of interference, designing a reliable vehicle attitude control system is a necessary condition to ensure that it moves in a predetermined trajectory. At present, PID control is widely used in attitude control, which is simple and reliable and stable in performance. However, in some harsh environments, the accuracy and stability of the vehicle attitude control are demanding, and it is difficult to meet the requirements of PID control technology^{[2][3]}. In order to solve this problem, the common methods are: fuzzy control, sliding mode control, dynamic inverse method and auto disturbance rejection control.

Sliding mode control system has the characteristics of anti-interference and anti-parameter perturbation^[4]. Although sliding mode control has strong robustness, it is easy to shake off the sliding mode surface. If there is great interference, the control precision is not satisfactory^{[5][6]}. The dynamic inverse method is dependent on the exact nonlinear model of the system, which is sensitive to the modeling error and can not deal with the unknown changes of the system. With the combination of dynamic inverse and neural network, the anti-interference ability is greatly improved, however, if the disturbance is larger, a certain degree of tracking error will be produced^[7]. The active disturbance rejection controller is suitable for solving the control problems with the existence of uncertainty, external interference and coupling^[8]. However, there are many parameters of the disturbance rejection controller, although some progress has been made in the parameter tuning problem, it has not been effectively solved^[9].

Fuzzy control is a kind of intelligent control method imitating human thinking. It can solve the control problem caused by inaccurate vehicle model and stochastic disturbance, and does not depend on accurate mathematical model. The fuzzy controller has strong anti-interference ability, fast response speed, and strong robustness to the change of system

parameters. However, Fuzzy control is difficult to solve the system's own steady-state error, PID control just to make up for this shortcoming^[10]. Therefore, in this paper, PID control and fuzzy control are combined in the missile control systems. This paper proposed the fuzzy PID control used on vehicle. In order to realize fuzzy-PID control, we find the fuzzy relation between three PID control parameters and the deviation signal, get deviation and the rate of deviation constantly, and then adjust the PID control parameters online according to fuzzy rules, to meet the requirements of different deviations and deviation rates. The simulation results show that fuzzy PID control can effectively accelerate the response speed, shorten the adjustment time, and the method can enhance the dynamic performance of the vehicle control systems.

2 FUZZY PID CONTROL

2.1 Attitude Control Model

Based on kinematic equations and dynamic equations, the vehicle guidance and control loop is established as Fig. 1.

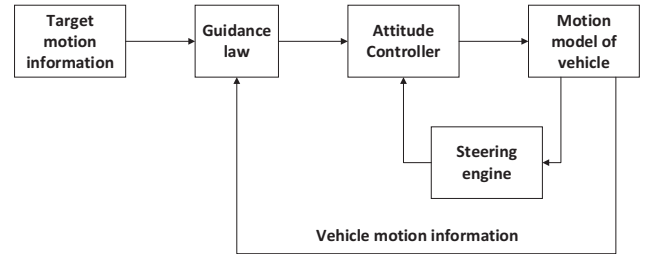


Fig 1. Vehicle Guidance and Control System

Fuzzy PID control is mainly used to control the attitude of vehicle. The following are the aerodynamic equations and kinematic equations of the pitch, yaw and roll channels^[11].

(a) Roll channel

$$\dot{\gamma} \approx \omega_x \quad (1)$$

$$\dot{\omega}_x = \frac{1}{J_x} (M_x^{\omega_x} \omega_x + M_x^{\delta_x} \delta_x) \quad (2)$$

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(b) Pitch channel

$$\dot{\alpha} = \omega_z - \frac{1}{mV} (c_y^\alpha \alpha - c_y^{\delta_y} \delta_y) \quad (3)$$

$$\dot{\omega}_z = \frac{1}{J_z} (M_z^{\omega_z} \omega_z + M_z^{\delta_z} \delta_z + M_z^\alpha \alpha) \quad (4)$$

$$n_y = \frac{c_y^\alpha}{mg} \alpha + \frac{c_y^{\delta_z}}{mg} \delta_z \quad (5)$$

(c) Yaw channel

$$\dot{\beta} = \omega_y + \frac{1}{mV} (c_z^\beta \beta + c_z^{\delta_y} \delta_y) \quad (6)$$

$$\dot{\omega}_y = \frac{1}{J_y} (M_y^{\omega_y} \omega_y + M_y^{\delta_y} \delta_y + M_y^\beta \beta) \quad (7)$$

$$n_z = \frac{1}{mg} (c_z^\beta \beta + c_z^{\delta_y} \delta_y) \quad (8)$$

The model of attitude control system is shown as Fig.2 to 4.

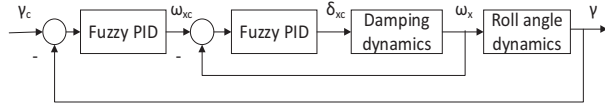


Fig.2. Roll Channel

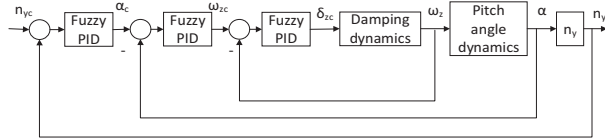


Fig.3. Pitch Channel

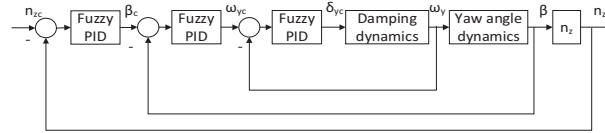


Fig.4. Yaw Channel

Where α is angle of attack, β is sideslip angle, γ is roll angle, $\omega_x, \omega_y, \omega_z$ are the projection of rotational angular velocity in aircraft body coordinate system, $\delta_x, \delta_y, \delta_z$ are rudder angles.

2.2 Fuzzy control

Fuzzy control uses programming language to summarize operator's control strategy, uses language variables and fuzzy set theory to form control algorithm.

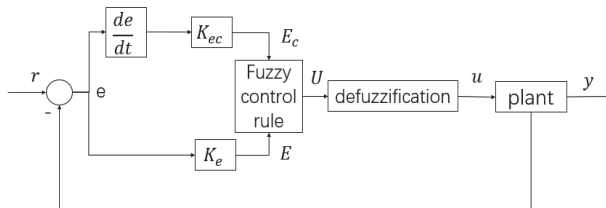


Fig.5. Fuzzy Controller

$$e(t) = r(t) - y(t) \quad (9)$$

$$ec(t) = \frac{de(t)}{dt} \quad (10)$$

Where $e(t)$ is deviation signal, $ec(t)$ is variation rate of deviation signal, E is the fuzzy quantity corresponding to $e(t)$, EC is the fuzzy quantity corresponding to $ec(t)$, U is the fuzzy control variable, u is the exact quantity corresponding to U .

The basic fuzzy controller has three main functions:

- Fuzzification: transform the exact quantity into the corresponding fuzzy quantity.
- Fuzzy inference: fuzzy inference according to the language rules (Fuzzy control rule table).
- Defuzzification: transform the control variable from the fuzzy quantity to the exact quantity that can be used for actual control.

2.3 Fuzzy PID Control

PID controller is one of the most commonly used control methods in automatic control of production process. However, due to the complexity of the actual control system and the precision of control, the traditional PID controller does not possess the function of online tuning of parameters, so it can not meet the requirement of parameter self-tuning under complex system parameter changing, and can not guarantee the stable control effect. A single fuzzy control can not satisfy the required precision.

The self-tuning PID controller realizes the purpose of setting control parameters by identifying the controlled process signals online, and its control effect depends on the accuracy of the identification model. In reality, despite the complexity of some systems, there are still many successful experiences in the control of complex systems. Fuzzy PID control is to save these successful experiences in the computer, and automatically adjust the PID parameters according to the actual situation detected by computers, so as to achieve real-time parameter tuning. In fuzzy control, the experience is transformed into fuzzy rule, and the conditions of fuzzy rule are obtained by fuzzing the sensor data, then fuzzy inference is applied, and the actual control parameters can be obtained after defuzzification.

In order to realize fuzzy PID control, first of all, find the fuzzy relation between three PID control parameters and the deviation signal. In the process of control, we get the rate of deviation and deviation constantly, and then adjust the PID control parameters online according to fuzzy rules, to meet the requirements of different deviations and deviation rates, so that the system can achieve good dynamic performance, static performance and robust performance.

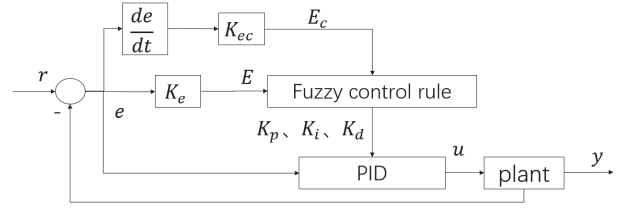


Fig.6. Fuzzy PID Controller

According to the basic characteristics of PID control, the requirements for controller parameters corresponding different $e(t)$ and $ec(t)$ are different^[12]:

- When the absolute value of $e(t)$ is very large, the value of K_p should be larger in order to eliminate the deviation and increase the response speed as soon as possible. In order to avoid overshoot at this time, the best value of K_i and K_d are zero.
- When the value of the deviation is small, in order to continue to eliminate the deviation and oscillation from prevent overshoot, the K_p should be reduced and the K_i can take the minimum value. The value of the k_d is determined according to the value of $ec(t)$.
- When $e(t)$ and $ec(t)$ are the same symbols, the controlled variables change toward the direction of deviating from the given value. When they are different, the controlled variable changes toward the direction approaching the given value. Therefore, when the controlled variable approaches a given value, the proportional action of the inverse number hinders the integral action, so that the integral overshoot and the consequent oscillation are avoided, when the controlled variable is far from the given value and is close to the given value direction, the control process will be slowed down due to the reverse of the integral and proportion. When the value of $e(t)$ is large, and the $ec(t)$ is negative, in order to accelerate the dynamic process of control, K_p should be negative.
- When the value of $ec(t)$ is large, K_p should take the minimum, K_i and should be a bigger value. Vice versa.
- The differential element is mainly used to control the deviation rate $ec(t)$, reduce overshoot and overcome the oscillation.

By using the basic principle and method of fuzzy mathematics, conditions and operation rules can be expressed by fuzzy sets, the fuzzy rules and the initial parameters of PID controller are saved in computer knowledge base, Then the computer can automatically adjust the PID parameters according to the actual response and fuzzy reasoning. The fuzzy rules for PID parameters can be summed up as Table 1 to Table 3.

When selecting linguistic variables, we should not only ensure the applicability and comprehensiveness of control rules, but also ensure its simplicity. In most cases, we choose PB(positive big), PM(positive medium), PS(positive small), ZO(zero), NB(negative big), NM(negative medium), NS(negative small) as language variable to construct fuzzy rules.

Table 1. Fuzzy Control Rule for K_p

E	EC						
	NB	NM	NS	ZO	PS	PM	PB
NB	PB	PB	PM	PM	PS	ZO	ZO
NM	PB	PB	PM	PS	PS	ZO	NS
NS	PM	PM	PM	PS	ZO	NS	NS
ZO	PM	PM	PS	ZO	NS	NM	NM
PS	PS	PS	ZO	NS	NS	NM	NM
PM	PS	ZO	NS	NM	NM	NM	NB

PB	ZO	ZO	NM	NM	NM	NB	NB
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Table 2. Fuzzy Control Rule for K_i

E	EC						
	NB	NM	NS	ZO	PS	PM	PB
NB	NB	NB	NM	NM	NS	ZO	ZO
NM	NB	NB	NM	NS	NS	ZO	ZO
NS	NB	NM	NS	NS	ZO	PS	PS
ZO	NM	NM	NS	ZO	PS	PM	PM
PS	NM	NS	ZO	PS	PS	PM	PB
PM	ZO	ZO	PS	PS	PM	PB	PB
PB	ZO	ZO	PS	PM	PM	PB	PB

Table 3. Fuzzy Control Rule for K_d

E	EC						
	NB	NM	NS	ZO	PS	PM	PB
NB	PS	NS	NB	NB	NB	NM	PS
NM	PS	NS	NB	NM	NM	NS	ZO
NS	ZO	NS	NM	NM	NS	NS	ZO
ZO	ZO	NS	NS	NS	NS	NS	ZO
PS	ZO	ZO	ZO	ZO	ZO	ZO	ZO
PM	PB	NS	PS	PS	PS	PS	PB
PB	PB	PM	PM	PM	PS	PS	PB

3 SIMULATION

In order to compare the performance of the tracking control instructions between the two methods, PID control and fuzzy PID control algorithm are simulated respectively.

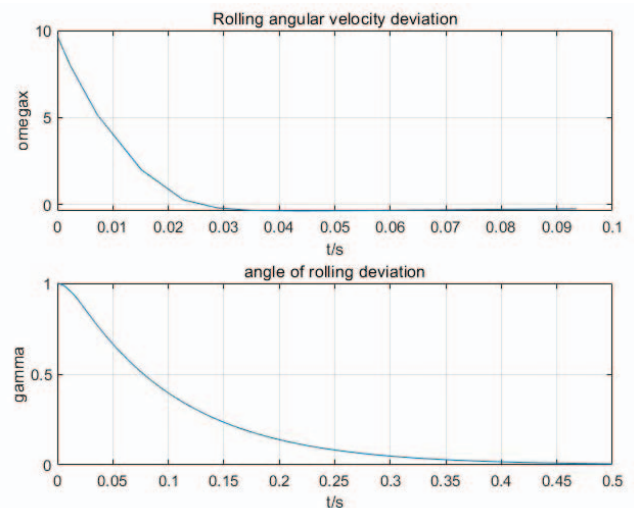


Fig 7. Fuzzy PID Control Algorithm on Roll Channel

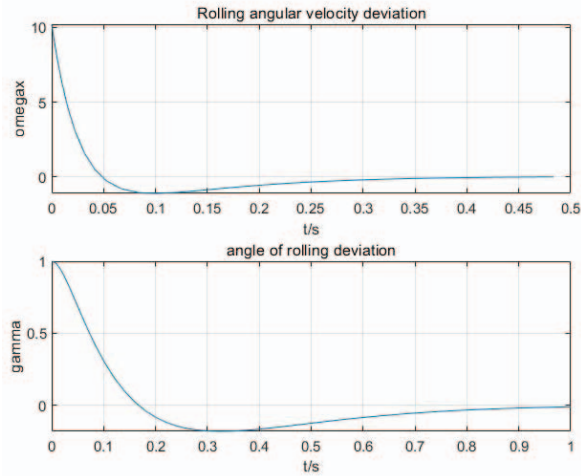


Fig 8. PID Control Algorithm on Roll Channel

It is visible from the figures, by using the traditional PID control method, the rolling angular velocity deviation can be reduced to zero in 0.35 seconds, and the fuzzy PID control method in 0.04 seconds. At the same time, the fuzzy-PID control vehicle can track roll angle instruction in 0.5 seconds, however, PID control vehicle needs 0.9 seconds.

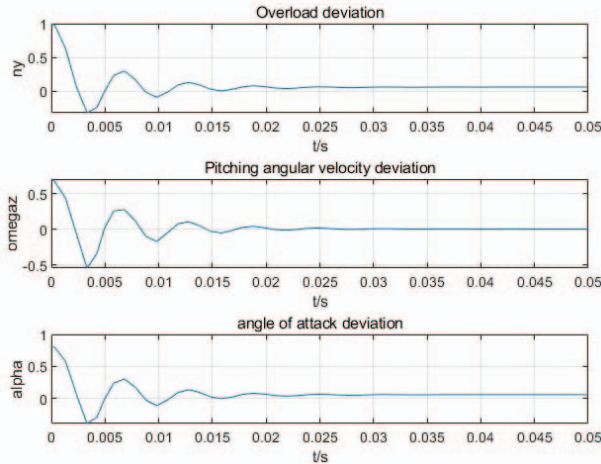


Fig 9. Fuzzy PID Control Algorithm on Pitch Channel

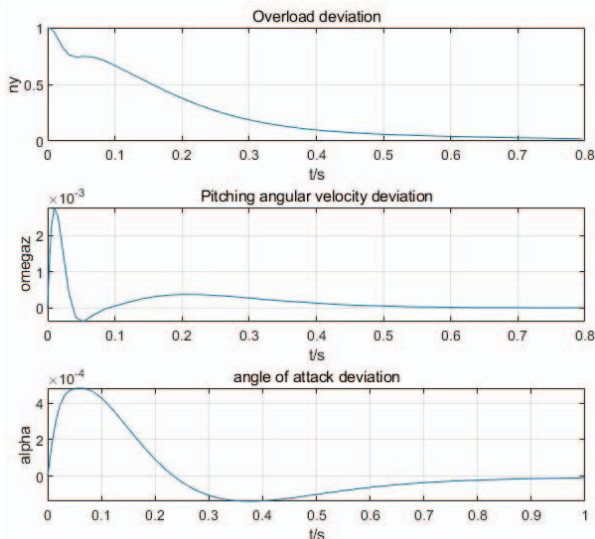


Fig 10. PID Control Algorithm on Pitch Channel

It is visible from the figures, by using the traditional PID control method, the pitching angular velocity deviation can be reduced to zero in 0.4 seconds, and the fuzzy PID control method use less than 0.02 seconds. At the same time, the fuzzy-PID control can track angle of attack instruction and overload n_y in 0.02 seconds, however, PID control needs 0.8 seconds.

In order to compare the effects of interference on Fuzzy-PID control and PID control, add a disturbance in gamma input with peak of 0.5 to the rolling channel at 0.01 second.

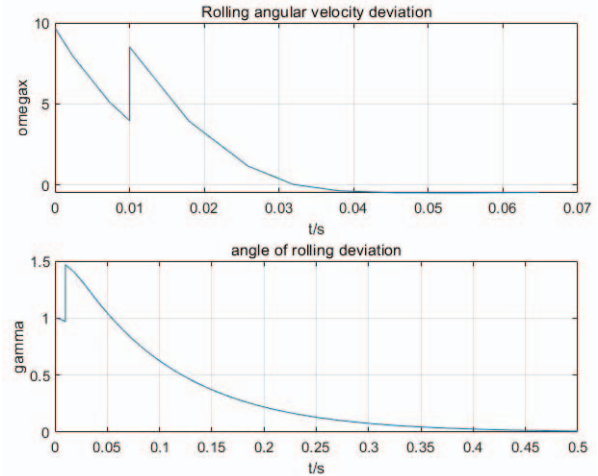


Fig 11. Fuzzy PID Control on Roll Channel with interference

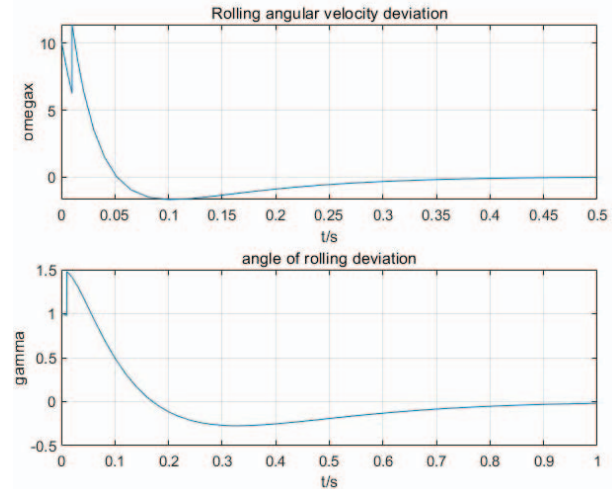


Fig 12. PID Control on Roll Channel with interference

It is visible from the figures, by using the traditional PID control method, rolling angular velocity deviation can be reduced to zero in 0.35 seconds, and the fuzzy PID control method use less than 0.04 seconds. At the same time, the fuzzy-PID control can track roll angle instruction in 0.04 seconds, however, PID control needs 0.8 seconds.

So the fuzzy-PID control can effectively accelerate the response speed.

4 CONCLUSION

In this paper, fuzzy PID control system is used to adjust vehicle attitude controller parameters automatically. From the simulation results, we can see that fuzzy-PID control effectively accelerate the response speed, the systems track overload commands and attitude angles more quickly, and stabilize aircraft attitude accurately. If there is an interference

in the system, the control system with fuzzy-PID responses much quickly and accurately than PID control. So fuzzy-PID control can enhance the dynamic performance of the vehicle control system.

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